

C:\Users\mthob\OneDrive\Documents\NetBeansProjects\FederatedIDSArtefact\src\main\java\com\registration\app\federatedid

```
1 /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3  * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4  */
5 package com.registration.app.federatedidsartefact;
6 import java.util.Random;
7 /**
8  *
9  * @author mthobi Local training - Each client represents an IoT gateway
10  * training on private local data
11  */
12 public class Client {
13
14     public Model localModel;
15     public double[][] X;
16     public int[] y;
17     public boolean isMalicious = false;
18
19     public Client(double[][] X, int[] y, boolean isMalicious) {
20         this.X = X;
21         this.y = y;
22         this.isMalicious = isMalicious;
23         this.localModel = new Model(X[0].length);
24     }
25
26     public Model trainLocalModel() {
27
28         if (isMalicious) {
29             System.out.println("⚠ Malicious client sending poisoned update...");
30             poisonModel();
31             return localModel;
32         }
33
34         localModel.train(X, y, 0.01, 10);
35         return localModel;
36     }
37
38     private void poisonModel() {
39         Random rand = new Random();
40         for (int i = 0; i < localModel.weights.length; i++) {
41             localModel.weights[i] = rand.nextDouble() * 10 - 5;
42         }
43         localModel.bias = rand.nextDouble() * 10 - 5;
44     }
45 }
```